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Chapter 14

# Reproductive options in mitochondrial disease

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## Abstract

Mitochondrial diseases require customized approaches for reproductive counseling, addressing differences in recurrence risks and reproductive options. The majority of mitochondrial diseases is caused by mutations in nuclear genes and segregate in a Mendelian way. Prenatal diagnosis (PND) or preimplantation genetic testing (PGT) are available to prevent the birth of another severely affected child. In at least 15%–25% of cases, mitochondrial diseases are caused by mitochondrial DNA (mtDNA) mutations, which can occur de novo (25%) or be maternally inherited. For de novo mtDNA mutations, the recurrence risk is low and PND can be offered for reassurance. For maternally inherited, heteroplasmic mtDNA mutations, the recurrence risk is often unpredictable, due to the mitochondrial bottleneck. PND for mtDNA mutations is technically possible, but often not applicable given limitations in predicting the phenotype. Another option for preventing the transmission of mtDNA diseases is PGT. Embryos with mutant load below the expression threshold are being transferred. Oocyte donation is another safe option to prevent the transmission of mtDNA disease to a future child for couples who reject PGT. Recently, mitochondrial replacement therapy (MRT) became available for clinical application as an alternative to prevent the transmission of heteroplasmic and homoplasmic mtDNA mutations.

## INTRODUCTION

New opportunities to prevent the transmission of mitochondrial diseases have emerged in recent years, based on novel insights in mitochondrial genetics and in occurrence and transmission of mitochondrial diseases, requiring guidance for their reproductive genetic counseling. For a patient and his or her parents, the very start is identifying the genetic defect that causes the mitochondrial disease in the patient, hereby not forgetting to remain critical whether the mutation is indeed explanatory. This may seem a "*sine qua non*"; however, in clinical practice, this is not always that obvious, as illustrated by a patient carrying a low percentage of the m.3243A>G mutation in combination with being compound heterozygous for *POLG* mutations and presumably by some more reported cases (Wong, 2007). Recurrence risk and preferred reproductive option(s) depend on multiple factors, the most important one being the primary genetic defect in the mitochondrial DNA (mtDNA) versus the nuclear DNA (nDNA).

## PREVENTING THE TRANSMISSION OF MUTATIONS IN THE NUCLEAR DNA

Although the mtDNA is entirely dedicated to the synthesis of 13 essential subunits of the respiratory chain,

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it does not by far provide all components necessary for mitochondrial function. Mitochondria also rely on an estimated number of 1000–1500 proteins, encoded by nuclear genes (Calvo et al., 2006; Pagliarini et al., 2008), which provide the majority of the OXPHOS components and play important roles in mtDNA replication and maintenance as well as in mitochondrial network dynamics. Obviously, there is a close and complex mito-nuclear interaction (Fig. 14.1).

Unlike the complex, maternal transmission of mtDNA mutations, discussed below, nDNA mutations implicated in mitochondrial disease follow Mendelian transmission patterns and are mostly autosomal recessively (AR) inherited or occur de novo (either X-linked or autosomal). Therefore, both prenatal diagnosis (PND) and preimplantation genetic testing (PGT) can be applied in a rather straightforward way. Obviously, such reproductive options are only possible when the causative genetic defect of the disorder concerned is known. Exactly that is the major challenge in mitochondrial diseases. Considering the huge number of potentially involved nuclear genes, of which more than 300 genes have already been implicated in mitochondrial diseases to date (Stenton and Prokisch, 2020), and the large and mostly not gene-specific variability in clinical and biochemical phenotypes, the difficulty in establishing a genetic diagnosis is not surprising. Identifying the genetic cause, enables parents to opt for reproductive testing options before (PGT), or in (PND), a subsequent pregnancy, although still a number of challenges exist.

## Monogenic versus multigenic phenotypes

When a couple with a child affected by a disease of unknown etiology, such as a mitochondrial(−like) disorder, ask for recurrence risk and potential options to prevent the birth of another affected child, the primary approach is to identify the cause of their child's disease. When the mtDNA is excluded, a targeted approach of genetic testing of individual nuclear genes fitting the phenotype could either result in (1) the establishment of a clear genetic cause (pathogenic variant), or (2) in the identification of a variant of unknown significance (VUS), or (3) in the inability to find genetic abnormalities at all. Only in the first situation reproductive testing options can be "simply" offered and merely the nature of the disorder in relation to the desirability to perform reproductive testing for, can be a point of discussion. For a VUS, reproductive testing is technically possible, but obviously not useful as long as its role in causing the disease is uncertain. At least a subclassification of pathogenicity is necessary to decide whether or not reproductive testing may be approved for suspicious variants in certain cases. Functional studies or the presence of mutations in the same gene in comparable patients may provide convincing evidence for the pathogenicity, but these are often lengthy or difficult to obtain in rare disease. For couples with a child wish this time factor is difficult; the context is different for parents who would like to know the cause of their child's disease, but do not have the desire for further children. For the majority of mitochondrial diseases reproductive testing has long not been possible due to the lack of a genetic defect. By performing next-generation sequencing (NGS) technologies such as whole exome sequencing (WES), a significant subset fortunately shifts to the first category. Notably, the more unbiased the genetic analysis is, the more relevant and challenging it becomes for clinical laboratory specialists and clinicians to critically evaluate whether an identified (likely) pathogenic mutation fits the phenotype. Besides solving more cases, which is the primary goal, also more VUS are being identified, not only in genes fitting the specific phenotype, but also in genes for which the association with the phenotype may be uncertain. This, as well as the still relatively time-consuming diagnostic process for WES in routine diagnostic setting, should be discussed with the couple so they can decide whether or not they wish to proceed or rather take their chances with a subsequent pregnancy. The severity of the disease in their child as well as the suspected mode of inheritance will likely influence their choice.

Not only for monogenic disease, the diagnostic yield is increased by WES. Also complex phenotypes consisting of multiple genetic defects, each responsible for a part of the phenotype, can be unraveled by this approach. Multigenic phenotypes in this context refer to diseases caused by multiple genetic defects which are individually categorized as "monogenic." Thus, patients have

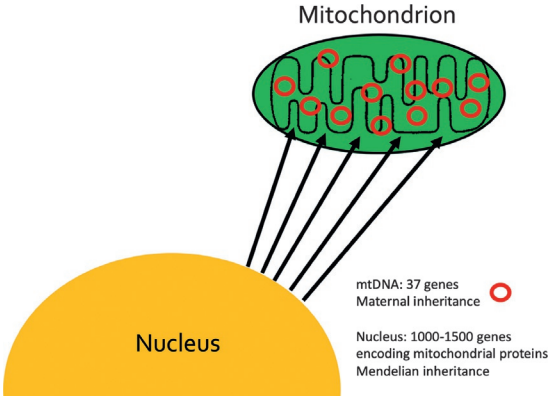



**Fig. 14.1.** Interaction between nDNA and mtDNA. The nucleus encodes 1000–1500 proteins which are imported in the mitochondrion and combined with the 13 proteins encoded by the mtDNA. The mtDNA is inherited maternally and the nuclear genes in a Mendelian way.

more than one pathogenic variant, which are unrelated to each other and would cause a phenotype when solely present as well, but together result in a mixed/complex phenotype. Although disease penetrance and expression may vary, with monogenic disease it is certain that an individual will not develop symptoms, if the pathogenic variant is absent. In other words, the pathogenic variant is an obligatory factor to develop disease. This contrasts with multifactorial disease where a single genetic defect will not result in a phenotype unless other, genetic or environmental, factors are present, and, conversely, the absence of the genetic variant may still result in the disease. For this reason, multifactorial diseases are not included in this chapter.

In multigenic disease, it is essential to determine, which clinical features can be attributed to which genetic defects and what the expected mode of inheritance is. The couple may have to decide for which inherited trait(s) they wish reproductive testing, considering both the severity of the diseases and the eventual chance of having an unaffected child, which reduces with the number of conditions tested for. It is also the question, which disorder warrants or permits reproductive testing, and this obviously also depends on the couple's perspective. When testing for multiple disorders in a PND setting, one could end up in a situation where the couple not just has to decide on whether or not to terminate the pregnancy for one disorder, but rather they have to make up their mind for different scenarios/combinations. Guidance of couples dealing with such difficult processes in a specialized setting is paramount. In such cases, PGT has an advantage to PND as the a priori frequency of an unaffected embryo can be low, but that embryo can be selected for by PGT. One should be aware though that an increase in the number of conditions tested also reduces the number of embryos suitable for transfer.

## WES: Informed consent and pretest counseling on incidental findings

A more unbiased genetic test, like WES, results in identifying more variants of uncertain significance (VUS) with respect to the phenotype. Also, variants causally involved with other phenotypes may be encountered. The extent to which this may happen depends on the approach of the data analysis. After performing WES, only genes implicated in a certain disease group (e.g., OXPHOS disorders, muscle disorders, movement disorders, etc.) may be analyzed resulting in findings that are expected to be at least related to the patient's phenotype. Identifying a causative mutation, which not only explains the phenotype present in the patient, but is also associated with an increased risk to develop other features, is possible as well. This is not unexpected, since the genes included in the panel are known before testing. This possibility is discussed with patients during pretest counseling, although the information that can be provided will be increasingly aspecific, when higher numbers of genes are tested. Such additional clinical manifestations associated with an established gene defect may influence a couple's considerations regarding reproductive testing of a subsequent pregnancy. In addition to the use of a virtual gene panel, the entire exome may be interrogated by a phenotype-driven approach, again focusing on features present in the patient. If the virtual panel includes "all known disease genes" ("Mendeliome") and/or analysis is not limited to a defined set of genes, there is a concern of detecting pathogenic variants not at all related to the phenotype, but to another disease. Different terminology is being used to refer to such findings that are unrelated to the reason for the test (e.g., unsolicited, unexpected, secondary or incidental findings). The latter seems to be the preferred term (Shkedi-Rafid et al., 2014), although appropriate terminology also depends on the way the finding was encountered in the data. Furthermore, a complete exome analysis may determine presumably pathogenic variants in genes with unknown function or disease association. Not only for the patient, but also in the context of recurrence risk and reproductive choices incidental findings may have consequences and should be considered.

Tiered consent has been proposed as strategy to deal with incidental findings (Bredenoord et al., 2011; Berg et al., 2011; Lolkema et al., 2013; Bijlsma et al., 2016). Tiered consent is differentiated consent, subdividing the potential findings in categories of which patients may, at least partly, decide whether or not they desire to receive them. Such categories may be based on (1) clinical consequences of the associated diseases, including for example the availability of preventive measures and/or treatment options (clinical utility), or the possibility of reproductive options (personal utility); (2) the risk to develop the associated diseases (clinical validity); and (3) the extent to which the clinical implications of variants are clear (pathogenicity). The exact nature and criteria of such categories are not easy to define. Regarding the pathogenicity, it is obvious that VUS are undesirable and should be redefined as either pathogenic or benign as soon as possible, necessitating adequate and up-to-date variant databases.

## Consanguineous versus nonconsanguineous families

The diagnostic yield of WES for mitochondrial disease varies among studies; numbers between 35% and 70% have been reported (Theunissen et al., 2018; Stenton and Prokisch, 2020). A significant subset of the patients

are consanguineous (Theunissen et al., 2018), which is unsurprising as mitochondrial disorders caused by nuclear gene defects are often autosomal recessive and consanguineous couples have an increased risk of offspring with autosomal recessive disease (Sallevelt et al., 2021). Consanguinity hampers the analysis because the homozygous regions are often large and include many potentially pathogenic variants. Moreover, the causative gene or variant is more likely to be very rare and therefore yet unknown to be associated with disease. Indeed, more VUS are identified in patients with consanguineous parents. Thus, it is more difficult to (firmly) establish the genetic cause in a subset of cases, which has consequences for the applicability of reproductive testing options.

Identifying causative mutations in consanguineous families not only provides recurrence risks and reproductive options for the family itself, but, for specific ethnic groups, it can also be of relevance for their countries of origin. In isolated subpopulations (Mathijssen et al., 2015) or subpopulations where consanguinity is custom, carrier frequency of rare variants can be high. There are global differences in the availability of (extensive) genetic testing, for example due to varying financial resources. When founder mutations are identified (Gerards et al., 2013) it is useful to register these in population-specific databases (Charoute et al., 2014) as this enables targeted testing, which is more widely available than WES, in the populations concerned. This way, a far greater number of couples can be informed about the specific risks for their subsequent offspring and may opt for reproductive testing. These benefits in our opinion outweigh the potential delicate issues of stigmas based on ethnicity.

## PREVENTING THE TRANSMISSION OF MUTATIONS IN THE MITOCHONDRIAL DNA (mtDNA)

In 15%–25% of cases, mitochondrial diseases are caused by mtDNA mutations. Two characteristics of mtDNA mutations, which are key to understanding the complexity of mtDNA disease transmission, include mitochondrial heteroplasmy and the genetic bottleneck. The majority of pathogenic mtDNA mutations resulting in severe disease are heteroplasmic, which means a mixture of mutant and wild-type mtDNA within a cell/tissue/individual. Heteroplasmy levels can vary between and within tissues of a carrier. Heteroplasmic mtDNA mutations are characterized by a threshold effect, meaning that there are no symptoms unless the mutant load (proportion of mutant mtDNA) exceeds a certain level. This threshold varies both within tissues and between different mutations and can depend on environmental factors, like the physicalcondition of the carrier. Transmission of mtDNA occurs only from females to their offspring and is subject to a so-called genetic bottleneck (Fig. 14.2). During oogenesis, the number of mtDNA molecules to be transmitted is reduced, and the resulting few mtDNAs become the founders for the offspring. In case the transmitting woman carries a heteroplasmic mtDNA mutation, this results in considerable variation in mtDNA

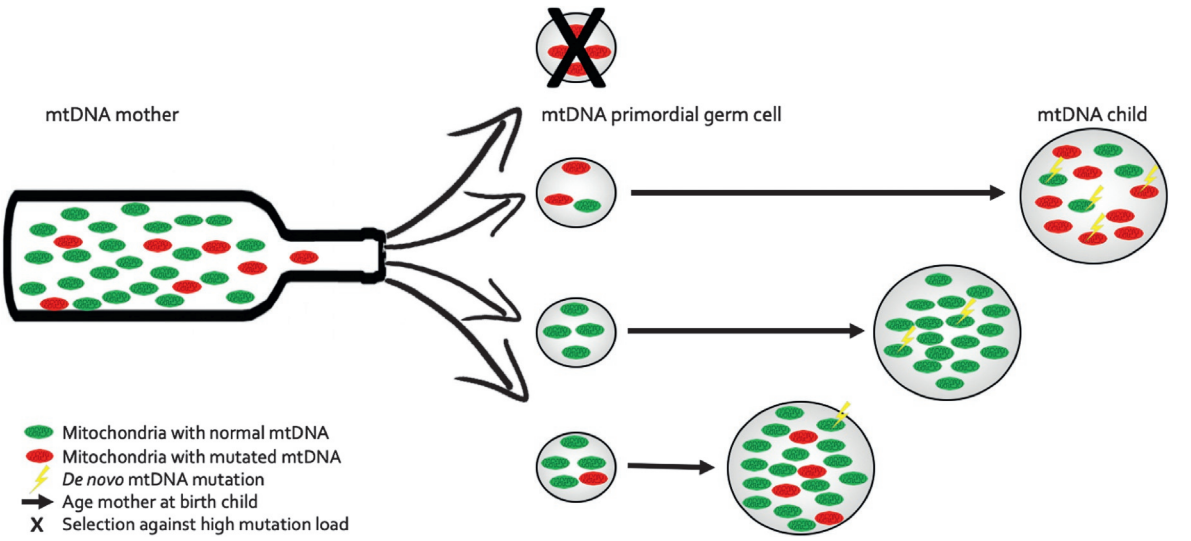



**Fig. 14.2.** The mitochondrial DNA bottleneck. During oogenesis, the number of mtDNA molecules to be transmitted is reduced, and the resulting few mtDNAs become the founders for the offspring. This results in considerable variation in mtDNA mutant load among her individual oocytes and subsequently among her offspring. Additional processes involved are genetic drift and mutation-specific additional neutral, negative or positive selection. Mutations increasingly occur de novo in aging oocytes.

mutant load among her individual oocytes (Howell et al., 1992) and subsequently among her offspring. Additional processes involved are genetic drift and mutation-specific additional neutral, negative or positive selection. The exact mechanism of the mitochondrial bottleneck is not fully understood, and some controversy exists concerning the content and "size" of the segregational unit (Lightowlers et al., 1997; Cao et al., 2007, 2009; Cree et al., 2008; Khrapko, 2008; Samuels et al., 2010). This size has been hypothesized to depend on the type of mtDNA mutation (Degoul et al., 1997; Blok et al., 1997; Brown et al., 2001; Monnot et al., 2011; Otten et al., 2018) and to be individual dependent for certain mutations (Monnot et al., 2011), possibly due to individual differences in initial mitochondrial copy number or genetic background. Irrespective of the mechanism, the smaller the lowest amount of remaining mtDNA (segregational unit), the more rapid a (complete)shift of the mtDNA genotype can occur.

To prevent the transmission of mtDNA disease, reproductive counseling is essential to discuss reproductive options related to recurrence risks (Fig. 14.3). Guidelines are formulated (Smeets et al., 2015; Poulton et al., 2019), but these are no black-and-white "prescriptions." For each couple, determining the most appropriate reproductive option, both from a medical and a personal point of view, will involve a process, including several decision points. The role of the counselor, not only in informing but also in structuring and sorting out what is most important for the couple, is crucial. Reproductive options that will be particularly addressed below are PND, PGT and Mitochondrial Replacement Therapy (MRT), also referred to as mitochondrial donation. Alternatively, accepting the risk for offspring without any testing is also an option for at risk couples. And, finally, oocyte donation and adoption are effective ways to prevent mtDNA disease transmissions well, but will not result in biologically-own children. These options will not be further discussed here, but it is relevant to mention that oocyte donation is not permitted in every country and that in countries where it is allowed legislation around anonymity of donors varies. The scarcity of egg donors, resulting in waiting lists, is another issue (which also applies to MRT).

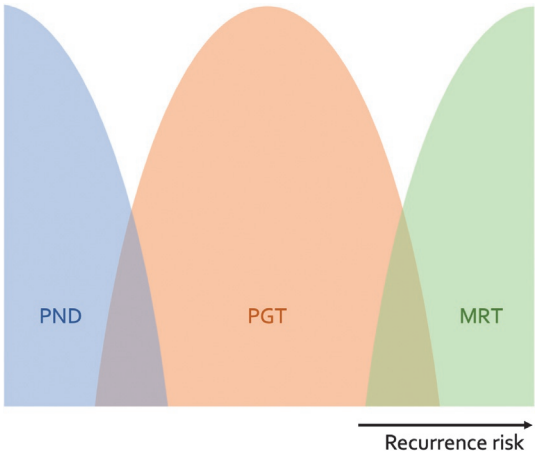



**Fig. 14.3.** Reproductive options for mtDNA mutations, related to recurrence risk (X-axis) and accuracy/burden/risks of the treatment. The recurrence risk depends on de novo versus familial occurrence, nature of the mutation and maternal mutation load (further discussed in text). PND, prenatal diagnosis; PGT, preimplantation genetic testing; MRT, mitochondrial replacement therapy.

## De novo mtDNA mutations

### De novo mtDNA point mutations

For de novo mtDNA point mutations, which represent about 25% of the mtDNA mutations and increase with age (Sallevelt et al., 2017b), the recurrence risk is low and probably close to 0%. However, it is hardly possible to establish de novo occurrence with complete certainty, and, mainly due to potential gonadal mosaicism in the mother, a recurrence risk of 0% cannot be provided. When the presence of the mutation in the mother has been excluded in several tissues, including a postmitotic tissue such as muscle or, for some mutations, urine (m.3243A>G; McDonnell et al., 2004), and with adequately sensitive techniques, no recurrence has been described. Based on recurrence cases from the literature where no postmitotic tissue in the mother was analyzed, and thus a mutation in the mother may have been missed, the recurrence risk is ~4% (Sallevelt et al., 2017b). This probably implies a lower actual recurrence risk in cases where the lack of detectable mutation levels is demonstrated in a postmitotic maternal tissue. Regarding reproductive counseling, this stresses the importance of including such tissues in evaluating de novo occurrence. For some specific mutations in particular (discussed in "Inherited point mutations" below), one should be aware that (near-)homoplasmic levels in an affected child may well result from very low mutation load in the mother. Thus, one should be careful in concluding that a mutation arose de novo. On the other hand, one should never base the recurrence risk on the (high) mutation load in the child without testing the mother, as is often done incorrectly in clinical practice. The test of the mother defines the recurrence risk.

Because of the low recurrence risk of (presumably) de novo mtDNA point mutations, prenatal diagnosis (PND) seems to be the preferred choice, if a couple wishes reproductive testing in a subsequent pregnancy. PND for mtDNA mutations has certain limitations, the most important one being the potential difficulties in interpreting the prenatal test result with respect to the phenotype

of the child. PND for de novo cases merely serves as reassurance, given the high likelihood that the mutation will be absent in prenatal samples in a subsequent pregnancy of the mother. An alternative reproductive option is preimplantation genetic testing (PGT). The invasive, lengthy and costly nature of this treatment, with generally reduced pregnancy rates compared with conceiving spontaneously, renders PGT a less obvious choice when recurrence risk is low. If the couple undergoes an assisted reproductive treatment (ART) for other reasons (e.g., fertility problems, nuclear gene defect) considerations could obviously be different.

#### Single large-scale mtDNA deletions

Other situations with low recurrence risk relate to single large-scale mtDNA deletions, which arise mostly de novo and consequently involve a low recurrence risk for mothers of affected children (Zeviani and Antozzi, 1997; Chinnery et al., 2004; Chinnery and Hudson, 2013). Also, for affected women harboring a single large-scale mtDNA deletion themselves, the recurrence risk is low. The latter has been established at ~1:24 (4.11%, CI 0.86%–11.54%) based on 40 women with an mtDNA disorder due to an mtDNA deletion and their 73 offspring, in which recurrence was reported in three (Chinnery et al., 2004). Therefore, also for single large-scale deletions, PND is the most suitable reproductive option.

## Inherited mtDNA point mutations

#### Low recurrence risk due to a low-level of an mtDNA point mutation in the female

Females carrying low heteroplasmy levels of mtDNA point mutations in postmitotic tissue generally have a low recurrence risk, but are a potentially challenging category regarding reproductive counseling, since the choice between PND and PGT may be difficult. Low recurrence risk and low mutation load are obviously linked to each other in this context, but not by definition. Mutation load is preferably established in multiple tissues, and again these ideally include muscle. "Low" mutation load could be arbitrarily defined as <10%, although this is not an evidence-based margin and may differ between mutations. As for the question "how to define low" for the maternal mutation load, determining a concrete recurrence risk for low-level carriers is difficult as well, as different mtDNA mutations have different transmission patterns (Otten et al., 2018).

### Prenatal diagnosis (PND)

Different segregation patterns exist for different mtDNA mutations due the combination of genetic drift with positive, neutral or negative selection. The skewed segregation of the m.8993T>G, for example, resulting from positive selection (Otten et al., 2018), means that the risk of affected offspring is higher for the m.8993T>G mutation than for other mtDNA mutations not positively selected for upon transmission. Nevertheless, carriers of the m.8993T>G mutation, particularly with low mutation load, are considered suitable PND candidates, because the correlation between mutation load and phenotype is rather good (White et al., 1999a), and mutation loads in oocytes/embryos show a distribution to the extremes; in the majority either 0% or ~100%. Due to this combination, prenatal samples are less likely to pose interpretation issues. Furthermore, recurrence risks can be quite adequately predicted based on maternal mutation load (White et al., 1999a).

A further prerequisite for reliable PND is the representativeness of the mutation load in the prenatal sample for the mutation load in postnatal tissues of the child. This includes stability of the mutation throughout embryofetal development and a relatively uniform distribution between fetal tissues, which has been shown for the m.8993T>G (and one m.8993T>C case) and m.9176T>C (one case) mutations, but, somewhat unexpectedly considering its postnatal behavior, also for m.3243A>G mutation (Harding et al., 1992; Suomalainen et al., 1993; Matthews et al., 1994; Jenuth et al., 1996; Ferlin et al., 1997; White et al., 1999a,b; Cardaioli et al., 2000; Dahl et al., 2000; Poulton and Marchington, 2000; Leshinsky-Silver et al., 2003; Bouchet et al., 2006; Steffann et al., 2006, 2007; Monnot et al., 2011). Although data are limited, this points to differences in segregation prenatally compared to postnatally, which might be related to the shorter prenatal time window, or to different demands of the metabolic system. Finally, the prenatal sample itself has to correspond to the fetal tissues, to be predictive. In line with this, should a couple opt for PND, the question comes up whether chorionic villus sampling (CVS) or amniocentesis is the preferred choice. CVS has the advantage of being performed earlier in the pregnancy, at around 11–12 weeks, enabling earlier termination in case of an unfavorable result. However, for mutations other than those at nucleotide position m.8993, data on representativeness of chorionic villi for fetal mutant load have been conflicting or insufficient. Some cases indicate analogy between CVS, amniocytes and fetal tissues for the m.3243A>G mutation (Bouchet et al., 2006; Monnot et al., 2011) However, m.3243A>G heteroplasmy differences of >10% in one study (Marchington et al., 2010) and even up to 45% in another (Vachin et al., 2018) were found within fetal placenta. Intra-placental mtDNA mutation load variations up to 55% have been reported without further details on the specific mutations and samples (Steffann et al., 2015), but have presumably been

reported later in more detail and include multiple mutations (Vachin et al., 2018). Significant variation has also been described between placenta and embryonic/postnatal tissues (Monnot et al., 2011; Vachin et al., 2018). This data might point to potential differences in mutation segregation between fetal tissues on the one hand and placenta on the other. Intra-placental variation particularly seems an issue when the placental sample is small ($\leq$10 mg or single cells) (Marchington et al., 2006, 2010; Monnot et al., 2011; Vachin et al., 2018), while the mean of the whole placenta seems to be comparable to amniotic and cord blood samples (Vachin et al., 2018). This is unsurprising considering the well-known variability in single cell mutation load in postnatal tissues, collectively resulting in a representative average mutant load upon testing. As 10 mg is a usual volume for CVS, the findings are relevant for clinical practice. Taken together, an amniocentesis in addition to CVS would be optimal for mtDNA mutations in order to collect the most reliable data for a diagnosis.

Similarly, one may consider to perform DNA-testing both on cultured and uncultured chorionic villi as long as data are lacking on potential heteroplasmy differences between both. For m.8993T>G/C such double testing (cultured/uncultured, CVS/amniocentesis) is probably not necessary given the relatively stable average mutant load between tissues and over time. Carrying out both CVS and amniocentesis has the drawback of necessitating two invasive procedures, each accompanied by a small risk of inducing a miscarriage, although recent studies indicate that this risk seems to be lower than usually quoted and may be negligible (Beta et al., 2019; Salomon et al., 2019). The need of performing an additional amniocentesis cannot entirely be predicted beforehand and therefore represents one of the decision points in the process. It depends, for example, on the mutation load identified in the CVS, and its proximity to the expression threshold. This threshold may be mutation-specific, but can only be accurately defined, if sufficient data is available, which is often not the case for rare or private mtDNA mutations. Nevertheless, a systematic review provides some guidance, showing that individuals with muscle mutant loads below ~18% are considered to be unaffected in >95% of cases, independent of the mtDNA mutation (Hellebrekers et al., 2012). This "generic threshold" was calculated after pooling data from all mtDNA mutations reported. For lower mutation loads, the likelihood that an individual will develop symptoms obviously further decreases (Hellebrekers et al., 2012). Clinical practices vary in the thresholds used for considering a fetus being likely affected or not. A rule of thumb where prenatal heteroplasmy <30% is associated with a positive postnatal outcome, while a value >60% would predict an affected individual, is also commonly applied (Vachin et al., 2018). For some mtDNA mutations this might be feasible, but it is a risky procedure, as for example, for the common m.3243A>G mutation, this increases the likelihood of being affected (with 95% confidence interval) to 20%.

If a mutation load identified with CVS is either very high or very low, additional amniocentesis may not be needed, since even with a significant error rate, expressed as increase or decrease relative to the identified heteroplasmy level, the mutation load will remain below or above the established threshold. This is corroborated by a study that found higher mutant load variance within the placenta, when the mean mutant load is closer to 50% (Vachin et al., 2018). The authors concluded that a mutant load value drawn from a CVS should be interpreted with caution, especially within the 20%–80% range, for m.3243A>G or m.9185T>C (Vachin et al., 2018). If a couple does not wish the risk of obtaining a second prenatal sample, they may prefer amniocentesis only, which seems a priori more appropriate than CVS as single test. This was also illustrated in the same study by good representativity of postpartum mutation load (Vachin et al., 2018). Still, even though by amniocentesis fetal cells from multiple organs are analyzed (epithelial cells from the urinary, respiratory, and gastrointestinal tracts, and from the epidermis), all these cells are dividing cell types, leading to differences in mutation load among daughter cells due to genetic drift. Therefore, combined with the paucity of available data, differential segregation cannot be excluded. Still, as genetic drift is random, the mean mutation load in all daughter cells combined may remain fairly constant, as long as sufficient cells are being used and a somatic bottleneck can be excluded. Furthermore, fresh amniocytes are believed to be favored over cultured amniocytes to avoid potential exposure to some selection of the mutant mtDNA (Vachin et al., 2018). An alternative approach is to keep a larger safety margin below the threshold for CVS, to take potential segregation differences between tissues and over time into account. Obviously, the lower the threshold, the safer it is, but at the same time the chance increases of terminating a pregnancy of a child that would not have been affected. The chance that the female produces offspring below the threshold in a subsequent attempt, should also be taken into account. When such an "extra safe" threshold is chosen, CVS may be regarded sufficiently reliable when the result is below this threshold. If slightly above, subsequent amniocentesis can still be considered.

## Preimplantation genetic testing (PGT)

An advantage of PGT for female carriers with a low-level mtDNA mutation load, irrespective of the mutation, is that mutation-free (or low-level) and thus transferrable

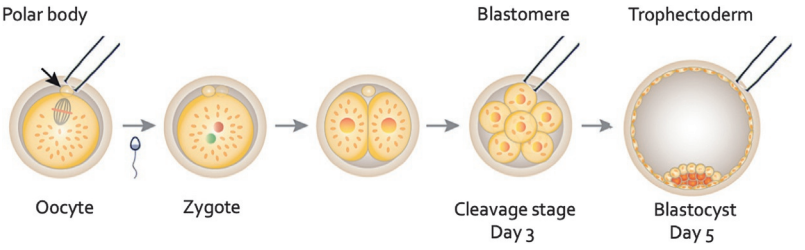



**Fig. 14.4.** Preimplantation genetic testing (PGT). During IVF, mature eggs are collected from the ovaries and fertilized by sperm in a lab. DNA is isolated from polar bodies (oocyte), a blastomere (cleavage state day 3) or trophectoderm cells (blastocyst day 5) to determine the mtDNA mutation load.

embryos are to be expected. Nevertheless, the earlier-mentioned disadvantages of PGT compared to spontaneous conception are also applicable. In all, making a choice between PND and PGT can be difficult for carriers with relatively low mutation load, depicted by an overlap in Fig. 14.3. Patient preferences play an important role as well; considerations regarding for example impact of terminating a pregnancy in case of a (likely) affected fetus and the burden of a PGT treatment are very personal. These are also likely to be influenced by a couple's personal history, for example, whether they have already had an affected child and if so, how severely affected it was. The same can be true for a carrier's own disease and/or family history. Furthermore, female's age and disease manifestations can both be a factor in eligibility for an ART (assisted reproductive technology)/PGT treatment.

### Moderate-to-high recurrence risk due to a higher-level of an mtDNA point mutation in the female

With increasing mtDNA mutation loads, recurrence risks become higher and often more difficult to predict. Obviously, a relation exists between the mutation load in the mother and the mutation load in her children, but many factors affect the eventual distribution in her oocytes, usually having variable mutation loads and consequently variable disease risks for each of the children.

## PGT: Blastomere biopsy in the cleavage-stage embryo

For carriers of heteroplasmic mtDNA mutations with moderate to high recurrence risks, which are the majority in clinical practice, we consider PGT as the best reproductive option available at present. It should be emphasized that PGT for mtDNA mutations is a strong risk reduction for having clinically affected offspring, albeit when no embryos without detectable mutation load are available and instead low-level embryos are transferred, it is not a complete risk exclusion. Notably, the policy and professional opinions regarding transfer differs between countries (Bredenoord et al., 2010b; Poulton and Bredenoord, 2010; Poulton et al., 2019). These differences, involving conservative and stringent transfer thresholds versus more liberal transfer policies, relate to the difficult balance between the highest chance of an unaffected child and the highest chance of a pregnancy. In most centers the genetic analysis is performed in one or two blastomeres derived from the cleavage stage embryo at day 3 postfertilization (Fig. 14.4). At this point in embryogenesis, the fertilized egg, containing between 50,000 and 550,000 copies of the mtDNA (Fragouli et al., 2015) has divided into 6–8 blastomeres. During cell division in the cleavage stage, the mtDNA is presumed to segregate randomly to daughter cells. Although some replication may occur during early preimplantation development in human embryos, it is yet unknown whether this takes place during the first 3 days (Monnot et al., 2013; Fragouli et al., 2015). Furthermore, available data suggest this to be limited to embryos with a mitochondrial deficit at this stage (Monnot et al., 2013; Fragouli et al., 2015) which seems corroborated by findings in pig embryos (Cagnone et al., 2016). Should this be true, then it means that mtDNA replication in the cleavage stage embryo is presumably not relevant in a PGT context, where embryos with absent or low-level mutations, and thus no mitochondrial failure, are selected. In bovine (May-Panloup et al., 2005) and mouse (Diot et al., 2016) embryos a (sharp) decrease in total mtDNA content between the 2 cell- and 4/8 cell-stage was shown, but this has not been reported in humans (Fragouli et al., 2015). Thus, for viable human embryos the total mtDNA amount of the embryo appears to remain largely constant. As a result, the mtDNA copy number per cell decreases upon cleavage. If a pathogenic mtDNA mutation is present, the mutation load per cell is expected to remain largely the same in the early embryo due to the high copy number, although small differences may occur by chance.

Variation in mutation load between cells has been reported to increase during later preimplantation development, as mtDNA amount per cell further decreases. This is illustrated in a mouse study, showing a strong mutation load correlation among individual blastomeres

of 2-, 4- and 8-cell stage embryos but with increasing 95% prediction intervals as the embryo progresses towards implantation (Neupane et al., 2014). Early evidence from human embryos showed that mutation loads are generally very consistent between blastomeres in most embryos, although (single) blastomeres can diverge in some (Steffann et al., 2006, 2014; Thorburn et al., 2009; Monnot et al., 2011; Vandewoestyne et al., 2012; Sallevelt et al., 2013). Based on these findings, the biopsy and analysis of two blastomeres was initially advised (Vandewoestyne et al., 2012; Sallevelt et al., 2013). The main drawback of analyzing two blastomeres instead of one is the decrease in live birth deliveries (De Vos et al., 2009), illustrating a dilemma between diagnostic certainty and pregnancy chances. In an extended dataset, a false negative error rate (i.e., classifying an embryo erroneously as below the threshold) of 0.34% could be confidentially calculated for the m.3243A>G mutation, based on a single blastomere analysis (Sallevelt et al., 2017a). For other mtDNA-mutations the numbers are (and will be) insufficient to draw this conclusion. However, the unbiased partitioning of mitochondria between daughter cells should make the m.3243A>G data also applicable to other mtDNA mutations, as the mutation itself does not seem to play a role at that stage. Taking account of this, biopsy of a single blastomere, rather than two, is now accepted practice, with favorable consequences for embryo viability and implantation rate.

There are situations, where in addition to the mtDNA defect testing for nuclear gene defects is requested in an embryo. Examples of such requests that we have encountered in clinical practice involved *POLG* and *BRCA2*. These days the preferred approach seems a subsequent trophectoderm biopsy to test for the nuclear gene defects, rather than biopsing a second blastomere. Couples have also requested the simultaneous selection for male gender, in order to prevent the transmission of even low-level mutation to a female fetus, who could be facing comparable dilemmas regarding reproductive options. This question is understandable; after all, selecting male offspring is an effective way to prevent transmission of mtDNA mutations to future generations. As such it is deemed morally acceptable (Bredenoord et al., 2010a), but currently not legally allowed in many countries. A potential drawback of the simultaneous selection for gender is the reduced number of remaining embryos eligible for transfer, thereby diminishing the chance of a successful PGT treatment for the couple. The consequences for future generations in case of transmission by female offspring have been modeled, showing that for a clinical threshold of 60%, the reduction of heteroplasmy levels to <5% dramatically reduces the chance of disease recurrence in subsequent generations, whereas transmitting >5% mutant mtDNA was associated with a significant risk of disease recurrence (Samuels et al., 2013). Notably, female offspring born after PGT and harboring low mutation levels will likely produce mutation-free embryos, which can be selected to fully clear the mtDNA mutation from the next generation. This is also true for preferentially-transmitted mutations that show skewing upon transmission. And finally, additional reproductive options will likely be available by the time the next generation is starting a family of their own.

## PGT: Trophectoderm biopsy in the blastocyst

A PGT technique that enables biopsy of a larger number of cells for genetic analysis is trophectoderm (TE) biopsy in the blastocyst stage at day 5 or 6 postfertilization (Fig. 14.4), has largely replaced blastomere biopsy in clinical practice for nuclear gene defects. In the blastocyst, comprising of a trophectoderm, inner cell mass and a fluid filled blastocoele cavity, the first cellular differentiation has occurred. The trophectoderm will develop into most of the fetal part of the placenta and the inner cell mass will later give rise to the fetus. At this point in embryogenesis, more cell divisions have occurred (the human blastocyst consists of >60 to >100 cells depending on the rate of cell divisions) and consequently the mtDNA content per cell is expected to have further decreased as long as mtDNA replication has not started. Although human data at this stage are lacking, animal studies indeed show a continued suppression of mtDNA replication in inner cell mass cells (St John, 2014), which are estimated to contain ~1000 mtDNA copies per cell (Shamsi et al., 2013). By contrast, evidence from studies in nonhuman (St John, 2014) and human (Hyslop et al., 2016) embryos indicates that mtDNA replication resumes in the trophectoderm before implantation, but mtDNA replication in inner cell mass cells is not believed to initiate before gastrulation (~ day 16 postfertilization) has taken place (Fig. 14.5).

Blastocyst-stage TE biopsy has several advantages, including less detrimental effects on embryo viability and implantation potential (Scott et al., 2013; Cimadomo et al., 2016). Also, the necessity for the embryos to develop to the blastocyst stage results in a further selection of the most viable embryos. However, because conventional culture media do not support extended blastocyst culture, it is necessary to vitrify biopsied blastocysts, which can also be detrimental (Cimadomo et al., 2016). That said, improved vitrification techniques and stimulation protocols seem to have resulted in equal cumulative live births and pregnancy rates between fresh and frozen embryo transfers (Zaat et al., 2021). For mtDNA mutations, a major advantage of TE biopsy would be a lower risk of missing potential differences in heteroplasmy levels

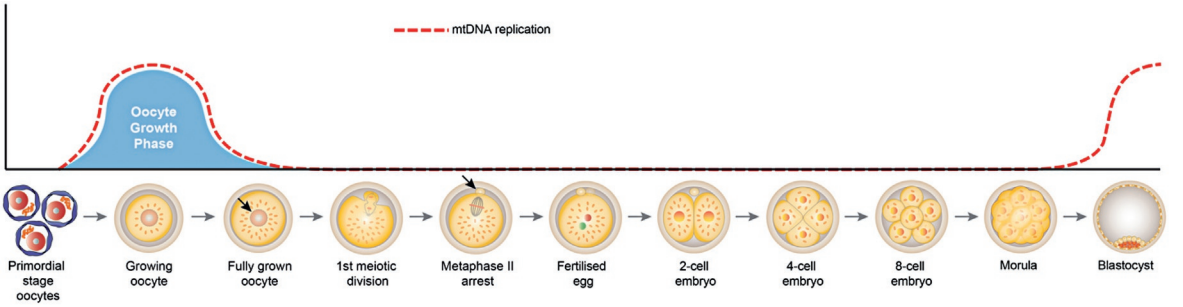



**Fig. 14.5.** mtDNA replication during oogenesis and preimplantation development. In mammals, the life time supply of oocytes is established during fetal development giving rise to a pool of relatively dormant primordial stage oocytes. Oocytes are recruited from this pool throughout life. Recruitment to growth is accompanied by upregulation of mitochondrial biogenesis and mtDNA replication. Fully grown human oocytes are estimated to contain ~500,00 copies of mtDNA. Following fertilization the embryo undergoes successive rounds of cell division in which the mtDNA is thought to segregate randomly between daughter cells without replicating. According to our current understanding, mtDNA replication is not initiated until shortly before implantation when the embryo has undergone blastocyst formation.

among cells, as multiple cells can be biopsied. However, paradoxically, heteroplasmy mosaicism is at the same time a potential drawback due to the mtDNA replication in the trophectoderm. So far, very limited human data exist on the representativeness of TE biopsy for the entire embryo and for the child with regard to mtDNA heteroplasmy levels. TE biopsy followed by PGT has been reported in two m.3243A>G mutation carriers (Treff et al., 2012; Heindryckx et al., 2014) (Table 14.1). In one of these, conflicting postpartum results have been reported in the resulting child (Treff et al., 2012; Mitalipov et al., 2014). Unfortunately, until now these discrepancies have not been explained by the authors, imposing a possible unjustified negative stamp on TE biopsy. Considering the timing of TE biopsy being closer to the lowest point of the bottleneck, blastomere biopsy in the cleavage-stage embryo seems preferable for PGT for mtDNA mutations. In support of this, it has been reported that intercellular variation in heteroplasmy levels increases during progression towards implantation in mice (Neupane et al., 2014).

## PGT: Implications of selection events and random drift?

An interesting aspect regarding PGT is the question whether selection events in the female germline upon transmission of mtDNA mutations might have implications for the reliability of PGT. The genetic analysis in the context of a PGT treatment is performed after the germline genetic bottleneck from mother to the embryo has taken place (Fig. 14.2), since this has happened during oogenesis in mother's own embryonic phase. Therefore, the PGT-results represent the mutation loads resulting from potential germline selection and are not expected to be altered by it anymore. Still, a "second bottleneck" could be expected in the embryo's somatic cells, since not only the cells destined to become germ cells undergo cell divisions postfertilization. Indeed, evidence for such a bottleneck comes from mouse (St John, 2014), monkey (Lee et al., 2012) and zebrafish (Otten et al., 2016) studies. Postimplantation, the mtDNA amount increases, whereby specialized cell types, once differentiated from pluripotent embryonic stem cells, acquire the appropriate mtDNA copy number to meet their needs for ATP production in a cell type specific manner (St John, 2014). Also in human fetal and postnatal tissues, varying and increasing mtDNA copy numbers have been demonstrated in different tissues (Steffann et al., 2015). This process may be regulated by epigenetic processes as suggested in mice (Kelly et al., 2012). In zebrafish, a vertebrate model convenient to investigate bottleneck events, the initiation of mtDNA replication seems to coincide with somitogenesis, when early organ development starts and cells shift from glycolysis to the use of oxidative phosphorylation (OXPHOS) (Otten et al., 2016). Interestingly, there is a partial parallel with the germ cells, as the metabolic shift observed in the human germline is coinciding with mtDNA replication: primordial germ cells are highly dependent on glycolysis (Ramalho-Santos et al., 2009), whereas developing oocytes, in which mtDNA content increases, rely on the OXPHOS for their ATP production (Van Blerkom et al., 1995; Dumollard et al., 2007).

Should the selection events, predicted to occur in the female germline (Otten et al., 2018), also take place during embryogenesis, PGT outcome might be influenced when positive selection for the mutation occurs, as is the case for the m.8993T>G mutation, due to increased mutation loads in the somatic tissues of the embryo/child. A selective advantage of the m.8993T>G mutation has been shown in somatic cell lines (Vergani

*Table 14.1*

**Published and unpublished children born after PGT for a pathogenic mtDNA mutation PGT because of de novo mtDNA mutation in previous child not included**

| | Embryo stage | Mutation | No. of embryos transferred | Embryo result (%) (../.. indicates result in two blastomeres) | Postpartum result (%) | Postpartum tissue(s) tested | Reference |
|---|---|---|---|---|---|---|---|
| 1. | Cleavage | m.8993T>G | 2 | 0/0 and 0/0 | 0 | CBl | Sallevelt et al. (2013) (Maastricht) |
| 2. | Cleavage | m.8993T>G | 1 | 0/0 | 0 | CBl | Unpublished (Maastricht) |
| 3. | Cleavage | m.8993T>G | 2 | 0/0 and 0/0 | 0 | CBl | Steffann et al. (2006) (Paris) |
| 4. | Cleavage | m.8993T>G | 1 | 2.5/2.5 | 3–5<br>5 | CBl<br>BM (4.5 yr) | Thorburn et al. (2009) (Melbourne) |
| 5. | Cleavage | m.8344A>G | 2 | 53 and 59 | 63 | ? | Steffann et al. (2014) (Paris) |
| 6. | Cleavage | m.83[a]A>G | 1 | 48 | n.d. | — | Unpublished/HFEA report* (Newcastle) |
| 7. | Cleavage | m.14487T>C | 2 | 5.4/6.1 and 12.2/12.3 | 5 | CBl | Sallevelt et al. (2017c) (Maastricht) |
| 8. | Cleavage | m.36[a]G>A | 2 | 5 and 2 | 7 | BM/U | Unpublished/HFEA report (Newcastle) |
| 9. | Cleavage | m.101[a]T>C | 1 | 1 | 1–2 | CBl | Unpublished/HFEA report (Newcastle) |
| 10. | Cleavage | m.130[a]T>C | 1 | 1 | 0 | CBl/Bl | Unpublished/HFEA report (Newcastle) |
| 11. | Cleavage | m.3243A>G | 1 | 12.6/13.4 | 16 | CBl | Unpublished (Maastricht) |
| 12. | Cleavage | m.3243A>G | 1 | 0 | n.d. | — | Unpublished (Maastricht) |
| 13. | Cleavage | m.3243A>G | 2 | 5/5 and 13/16 | 5 | CBl | Monnot et al. (2011) (Paris) |
| 14. | Blastocyst | m.3243A>G | 1 | 12 | 15<br>0<br>10<br>47/52<br>46/42 | BM (1 mo)<br>Same lab as PGT<br>BM/U (5 and 12 mo) “commercial lab”<br>Bl (5 and 12 mo) “commercial lab”<br>Bl/U (1.5 mo)<br>Bl/U (18 mo) Different lab | Treff et al. (2012) (New Jersey)<br>Mitalipov et al. (2014) |
| 15. | Blastocyst | m.3243A>G | 1 | 0/0 | 0 | CBl/S/U | Heindryckx et al. (2014) (Ghent) |

[a]Characters hidden to respect confidentiality.

CBl, cord blood; BM, buccal mucosa; U, urine; Bl, peripheral blood; S, saliva.

et al., 1999), which implies this is not a germline-specific process. Important to note is that the initial m.8993T>G heteroplasmy levels in this study were high (85%–95%). The potential selective advantage corroborates another study, showing an increase of 0.75% per year in blood of an individual with high (79%) m.8993T>G mutation load, whereas in an individual with low (7%) mutation load this percentage remained stable over 23 years (White et al., 1999c). Although generally no significant differential segregation throughout embryofetal development is reported at varying heteroplasmy levels (Harding et al., 1992; Bartley et al., 1996; Ferlin et al., 1997; Steffann et al., 2007), at a closer look, three cases with somewhat larger differences between PND samples and fetal tissues are in the higher mutant load range (Harding et al., 1992; Steffann et al., 2007). Whether these differences indeed result from preferential segregation or merely from random drift is unclear. The only embryo transferred after PGT that contained the m.8993T>G mutation harbored 2.5% mutation load in the blastomere(s) and 4% in postpartum cord blood (Thorburn et al., 2009) (Table 14.1). Taken together, as in the germline (Otten et al., 2018), in somatic tissues positive selection seems to be a relevant factor at high, but less at low mutation loads. Conveniently, for this mutation often mutation-free embryos are available due to the skewed segregation.

In contrast, negative selection occurs for the m.3243A>G mutations, where mutation loads >80%–90% were never observed in blastomeres, implying that these mutation loads seem incompatible with oogenesis and are selected against (Otten et al., 2018). The net result for PGT is that more embryos have a lower mutation load than would be expected by genetic drift, based on the mutation load of the mother. This would increase the number of embryos below the threshold of expression slightly. Not only selection processes, also random genetic drift has to be taken into account and for this too the mutation load of the transferred embryo is of importance. An increase in average mutation load may by chance occur upon random genetic drift. Both positive selection and genetic drift should be evaluated when establishing a threshold for transfer, instead of merely relying on clinical expression in relation to mutation load. The generic threshold of 18% seems safe, but caution is warranted, if a higher threshold is considered for a given mtDNA mutation. This is illustrated by an m.3243A>G embryo that showed a relative increase of ~20% when comparing the PGT and postnatal results (~13% and 16% heteroplasmy, respectively, Table 14.1). Mutation loads above the threshold are at increased risk of segregating to clinically significant levels. The lower the initial mutation load is, the higher the chances are that it remains stable upon random drift or, more likely, disappears pre- or postnatally. Although data are yet limited, stable segregation is observed in the other cleavage-stage m.3243A>G and the m.14487T>C PGT results (Table 14.1). With the current progressed insight, a transfer threshold of 60% which we considered appropriate for the m.9176T>C mutation (unpublished data), based on a high (>90%) phenotypic threshold, should be critically re-evaluated. Given the possible skewed pattern of this mutation (Jacobs et al., 2005, 2007; Hung and Wang, 2007), it is advisable to investigate whether in addition to genetic drift positive selection is involved. In fact, each mtDNA mutation should ideally be evaluated this way.

## PGT: Further prerequisites for successful outcome

The success of PGT depends on the availability of embryos with either 0% or below-threshold mutation loads. We are aware of ~90 PGD cycles performed in ~50 mtDNA mutation carriers worldwide (published data from Paris (France) (Steffann et al., 2006, 2014; Monnot et al., 2011), Maastricht (The Netherlands) (Sallevelt et al., 2013, 2017a,c), Ghent (Belgium) (Vandewoestyne et al., 2012; Heindryckx et al., 2014), Melbourne (Australia) (Thorburn et al., 2009) and New Jersey (USA) (Treff et al., 2012) and unpublished data from Newcastle, Paris and Maastricht)). Fifteen babies born after PGT performed in mtDNA mutation carriers are listed in Table 14.1. The chance of mutation-free or low-level mutation embryos is in general inversely correlated with the mutation load of the carrier, depending on genetic drift and additionally on the segregation pattern of the mtDNA mutation. This can be either a more diverse (m.3243A>G) or more compact (m.8344A>G) distribution pattern or a role for selection mechanisms. When the mutated mtDNAs are preferentially transmitted, similarly to the m.8993T>G mutation (Otten et al., 2018), a female with high heteroplasmy levels is less likely to have transferable embryos. Interestingly, for the m.9176T>G mutation an increased mitochondrial membrane potential (MMP), which is the suggested mechanism underlying the positive selection of the m.8993T>G mutation (Otten et al., 2018), has been demonstrated as well (Carrozzo et al., 2001). Although only two families with this mutation have been reported to date, it is striking that in both a healthy carrier mother with ~50% mutation load in blood conceived two children affected by Leigh syndrome resulting from (near-)homoplasmic levels (Carrozzo et al., 2001; Akagi et al., 2002), which may indeed be compatible with preferential transmission of mutated mtDNAs. Given the moderate mutation load of this particular mother, she is expected to have oocytes with low mutation load as

well, "escaping" the (potential) positive selection, and thus she could benefit from PGT.

The maternal mutation load at which not enough suitable embryos for transfer are produced, is not a black-and-white threshold, it is a chance event with many players involved. Consequently, a female carrier may not be able to produce sufficient embryos, which develop to good quality blastocysts, i.e., capable of developing to a stage where it has the potential to implant, eligible for transfer in a PGT setting. In the case of the common pathogenic variant m.3243A>G mutation, a model based on pedigree analysis has been developed to predict the probability of obtaining embryos with sufficiently low variant loads by PGT (Pickett et al., 2019). This model is being refined by the development of an algorithm (Lindsey et al., unpublished data), which calculates the likelihood of having an unaffected child by PGT, taking not only into account the mtDNA mutation load of the carrier, but also the distribution pattern of the mtDNA mutation upon transmission (genetic drift/positive/negative selection) and the number of oocytes produced per cycle. Obviously, the higher number of embryos available, the higher the chance of achieving a pregnancy with a high quality embryo. PGT seems acceptable to offer, if a cycle is predicted to yield at least one embryo below the threshold of expression, even though the quality and subsequent likelihood of a pregnancy cannot be calculated upfront. Based on the experience during the first cycle, the chance of having an unaffected child can be recalculated. The final determinant in the outcome of PGT is the pregnancy rate. Unfortunately, not every embryo transfer results in a viable pregnancy. Chances are on average 30% per cycle (Theobald et al., 2020), depending on age and other known and unknown factors. With respect to legislation, one should be aware that PGT may not, or not for the same conditions, be allowed in every country. Furthermore, different funding schemes among countries result in varying accessibility of PGT for individual couples.

#### High recurrence risk due to a (very-)high-level to homoplasmic mtDNA point mutation in the female

Carriers of very high mutation loads themselves (such as in our series an m.9176T>C carrier with ≥90% mutant level, unpublished data) or of specific mutations with compact distribution patterns (m.8344A>G) are unlikely to produce embryos with mutation loads below the expression threshold. Homoplasmic mtDNA mutation carriers constitute another category for whom PGT by definition is not able to decrease or exclude mutation levels transmitted to the offspring, since all embryos will harbor 100% mutant load. The most prevalent and well-known homoplasmic mtDNA mutations are associated with Leber hereditary optic neuropathy (LHON), resulting in severe vision loss (Yu-Wai-Man and Chinnery, 1993). There is a reduced penetrance which is gender-dependent, with female carriers having on average ~10% risk of developing LHON and male carriers ~40% (Yu-Wai-Man and Chinnery, 1993). Accordingly, each pregnancy of a homoplasmic LHON mutation carrier has a risk of ~20% of a son with vision loss and a ~5% risk of a daughter with vision loss. In our experience, a handful of couples has requested sex-selective PGT with transfer of girl embryos in order to reduce the risk of an affected child. Although this may pose some ethical questions, PGT for gender selection in these cases is generally deemed morally acceptable (Bredenoord et al., 2008; Poulton and Bredenoord, 2010), as for long it has been the only risk-reducing strategy towards offspring available for homoplasmic LHON carriers. When there is another reason necessitating an IVF treatment, the "proportionality," in this context referring to the (additional) treatment risks and burden in relation to the achieved risk reduction, is more balanced compared to couples without IVF indication. Still, the views of the couples themselves can be very personal and variable, whereby for example differences in the perceived severity of the disease, heaviness of the treatment and/or worthiness of the risk reduction play a role. More importantly, despite general ethical consensus, legislation may not allow PGT for this indication.

### Mitochondrial replacement therapy (MRT)

Like PGT, MRT involves the application of assisted reproductive technologies (ART) to reduce the risk of transmission of mtDNA disease. Conceptually, MRT involves rescuing the nuclear genome from its defective mitochondrial environment by transplanting it to an enucleated egg or zygote from an unaffected donor. This offers the potential to enable an affected woman to have a biologically-own child with a greatly reduced risk of transmitting mtDNA disease. MRT has recently been translated to clinical treatment for women at risk of transmitting serious mitochondrial disease to their children. However, the technology is still in its infancy and is currently in the UK offered only to women who would be unlikely to benefit from PGT. Such cases include those who carry a pathogenic variant at homoplasmic levels or at very high levels of heteroplasmy (reviewed in: Greenfield et al., 2017).

Nuclear genome transplantation for MRT can be performed either before or after the egg is fertilized. Progression through oocyte maturation and fertilization provides a number of opportunities for microsurgical extraction of the nuclear genome. In fully-grown immature oocytes, the nuclear genome is contained in a large nucleus known

as the germinal vesicle (GV). The diploid genome of the immature oocytes is halved to generate the single copy maternal genome during two successive meiotic divisions. Both meiotic divisions are highly asymmetrical, minimizing the loss of oocyte cytoplasm by ejecting half of the chromosomes in a nonviable polar body. The first meiotic division is completed before the mature egg is ovulated, entering a state of arrest at metaphase of meiosis II until the egg is fertilized. The second meiotic division, triggered by a sperm entry, results in two single copy genomes, one of which is lost to the second polar body. During fertilization, the haploid maternal genome retained in the egg and the haploid paternal genome introduced via the sperm, become separately enclosed in pronuclei, which enlarge and migrate towards each other during progression through the first zygotic cell cycle (reviewed in Herbert et al., 2015).

Since the efficacy of MRT depends on the production of viable embryos with a low level of maternal mtDNA, the optimal stage for transplantation is determined by these two parameters. In theory, it should be possible to transplant the oocyte nuclear genome at the GV stage. However, studies in mouse indicate poor embryo development following GV transfer and there have been no reports of successful GV transfer between human oocytes (Darbandi et al., 2017). By contrast, a number of studies indicate that transplantation of the nuclear genome before fertilization when the egg is arrested at metaphase of meiosis II, or after fertilization when the pronuclei have formed, provide promising approaches to MRT (Fig. 14.6, reviewed in Greenfield et al., 2017). In an interesting development, the possibility of using the genomic DNA ejected in the polar bodies has been tested in the mouse (Wang et al., 2014; Ma et al., 2017a). An advantage would be that the by-products of female meiosis could be salvaged to provide additional maternal genomes for use in MRT. The findings from studies in the mouse indicate that fusion of the first polar body with enucleated metaphase II eggs resulted in efficient blastocyst formation and the birth of live pups. Second polar body fusion with enucleated zygotes was somewhat less efficient (Wang et al., 2014; Ma et al., 2017a). Whether these findings can be readily translated to human is unclear. Based on the limited evidence, embryo development was poor following fusion of the first polar body with metaphase II-arrested human eggs (Ma et al., 2017a).

Because pronuclear transfer (PNT) and meiosis II spindle transfer (MST) are currently the most prominent approaches to MRT (Fig. 14.6), we will cover these techniques in more detail. The technique of PNT was pioneered in the mouse almost four decades ago by McGrath and Solter, who reported that PNT between zygotes from different mouse strains resulted in healthy and normally-reproducing offspring (McGrath and Solter, 1984). PNT was subsequently used to investigate mtDNA segregation during early embryonic development in the mouse (Meirelles and Smith, 1997, 1998). Later experiments using zygotes from a mouse carrying a rearrangement in the mtDNA indicated that the fraction of mutant mtDNA could be greatly reduced by transplantation of pronuclei into enucleated zygotes from females with wild-type mtDNA (Sato et al., 2005). The PNT procedure developed in the mouse involved treatment of zygotes with microtubule and actin-depolymerizing drugs. This facilitates enucleation by increasing the fluidity of the zygote cytoplasm. The enucleation procedure involves extraction of pronuclei in a karyoplast containing a small amount of cytoplasm bounded by a fragment of the zygote plasma membrane. Fusion of the karyoplast with the enucleated zygote was induced either by using inactivated Sendai virus or an electrical pulse (McGrath et al., 2017).

The first proof of concept for PNT to prevent transmission of mtDNA disease in human zygotes was published in 2010 (Craven et al., 2010). These studies were conducted using abnormally fertilized human zygotes donated by couples undergoing ART treatment for infertility. Using this material, it was possible to demonstrate technical feasibility of PNT in human zygotes following some technical modification of the approach used in earlier studies in the mouse (Craven et al., 2010). Having established technical feasibility of PNT in human zygotes, the next step was to determine whether it was compatible with development to the blastocyst stage. Because abnormally fertilized eggs do not develop to the blastocyst stage with high efficiency, it was necessary to use eggs fertilized specifically for research. Initial results indicated poor survival and onward development of those that fertilized normally, when using procedures optimized in abnormally fertilized zygotes (Hyslop et al., 2016). It was postulated that this might be due to accelerated progression of the latter through the first zygotic division leaving insufficient recovery time between the PNT procedure and division to the 2-cell stage. Consistent with this, survival was dramatically improved by transplanting the pronuclei shortly after they became clearly visible (~8 h after sperm injection) rather than shortly before they disappeared (16–18 h after sperm injection) (Hyslop et al., 2016). Following further optimization of the procedure, it was reported that 40% of PNT embryos developed to the blastocyst stage compared with 60% of unmanipulated control zygotes (Hyslop et al., 2016). By contrast, blastocyst formation following removal and replacement of pronuclei in the same zygote was equivalent to unmanipulated controls (Hyslop et al., 2016). Ongoing work aims to uncover the causes of reduced blastocyst

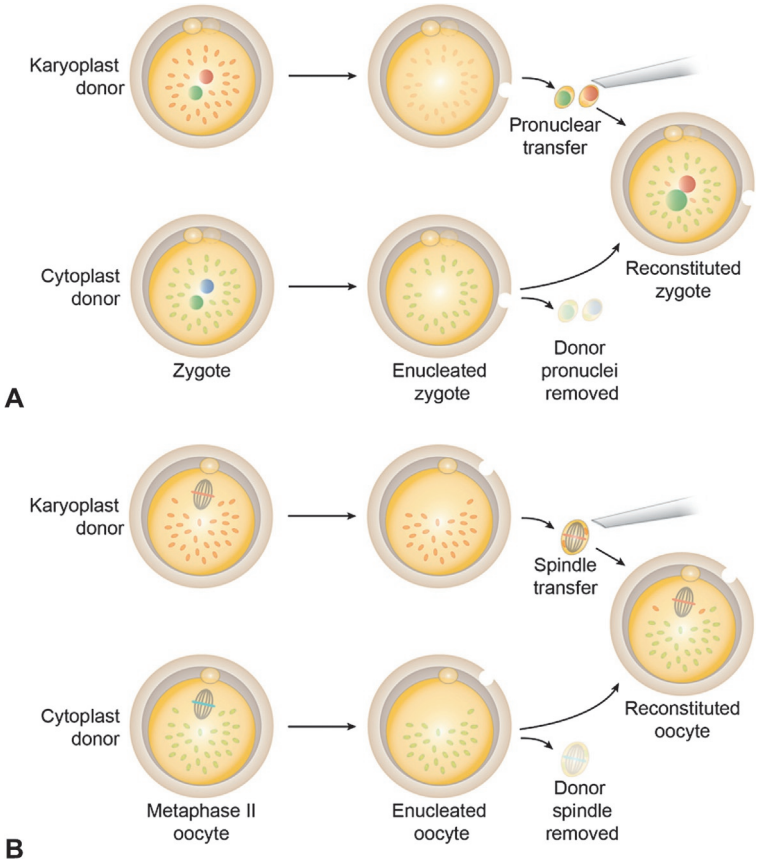



**Fig. 14.6.** Mitochondrial replacement therapy. Panel A displays pronuclear transfer (PNT) between an mtDNA carrier-derived zygote and a donor zygote. Panel B displays meiosis II spindle transfer (MST) between an mtDNA mutation carrier-derived oocyte and a donor oocyte.

formation following PNT (see below). A more recently reported variant of the PNT procedure, known as prepronuclear transfer, involves extraction of the maternal genome during the second meiotic division (Wu et al., 2017). This can be done without the use of cytoskeletal inhibitors and involves transplantation of only the female pronuclei (Wu et al., 2017). The experiments were performed using in vitro matures oocytes, which have reduced potential for blastocyst formation, nonetheless, the results appear promising (Wu et al., 2017) and further investigation of this approach is warranted.

A number of research groups have investigated the potential of meiosis II spindle transfer (MST) for preventing transmission of mtDNA disease (Greenfield et al., 2017). As described above, the oocyte chromosomes are normally aligned on the meiosis II spindle at this stage and cannot be visualized by brightfield microscopy. However, progress in MST has been greatly enhanced by the use of liquid crystal birefringence to visualize the spindle microtubules (Oldenbourg et al., 1998). The meiosis II spindle and the associated chromosomes are extracted in a karyoplast in the presence of an inhibitor of the actin cytoskeleton. Fusion of the karyoplast with an enucleated metaphase II egg can be achieved using inactivated Sendai virus (Tachibana et al., 2010). Proof of concept studies in rhesus macaque eggs indicated fertilization of eggs following MST resulted in viable embryos and the birth of healthy monkeys (Tachibana et al., 2009). Subsequent studies indicated that MST is also feasible in human oocytes (Tachibana et al., 2013; Paull et al., 2013). However, in contrast to the monkey, a high incidence (48%) of abnormal fertilization was observed following MST in the human (Tachibana et al., 2013). Nevertheless, those that underwent normal fertilization developed to the blastocyst stage with reasonable efficiency (Tachibana et al., 2013). While the high incidence of abnormal fertilization is a clear limitation of MST, it may be possible to overcome this through further refinement of the technique. Indeed, a subsequent report of a live birth following MST for prevention of mtDNA disease indicates that it is compatible with development to term in humans (Zhang et al., 2017).

While the efficiency of embryo development to the blastocyst stage is a primary determinant of success of MRT, preventing transmission of mtDNA disease crucially depends on minimizing the contribution of the maternal mtDNA to the mitochondrial genome of the embryo. This consideration arises from the fact that transplantation of the nuclear genome involves co-transfer of a small amount of cytoplasm, which generally contains some mitochondria. Co-transfer (also known as carryover) of mtDNA with the nuclear genome can result in heteroplasmy for maternal mtDNA in the embryo. Thus, nuclear genome transplantation from eggs with high levels of pathogenic mtDNA variants could result in transmission of a considerable fraction of mutation-carrying mtDNA. Carryover of mtDNA during PNT in human zygotes was reduced through a series of optimizations resulting in heteroplasmy levels ranging from undetectable to less than 5% (Hyslop et al., 2016). Heteroplasmy levels following MST in human oocytes were reported to be <2% (Tachibana et al., 2013; Paull et al., 2013). To test the fate of co-transferred maternal mtDNA, embryonic stem (ES) cell lines derived from PNT and MST embryos, were monitored for shifts in heteroplasmy during proliferation in vitro. Remarkably, it was found that some ES cell lines derived from PNT and MST blastocysts showed complete "reversion" to the maternal mtDNA during proliferation in vitro (Yamada et al., 2016; Hyslop et al., 2016; Ma et al., 2017b). While the majority of studies were conducted using eggs donated by women who did not carry pathogenic mtDNA variants, homoplasmy for maternal mtDNA in ES cells was observed in one study involving egg donors carrying pathogenic mutations (Hudson et al., 2019). While it is unclear whether reversion reflects a quirk of mtDNA segregation in ES cells, these unexpected findings raised the possibility that MRT might not be effective in preventing transmission of mtDNA disease in all cases.

How such a small fraction (2%–5%) of mtDNA outcompetes the mtDNA contained in the cytoplasm of the enucleated egg remains mysterious. It has been suggested that mtDNA reversion may be driven by sequences in the noncoding region of the mtDNA D-loop, known as the conserved block II (CSBII) (Ma et al., 2017b). The CSBII region is a transcription termination site at which mtDNA replication primers are generated. It consists of a single adenine residue flanked by two runs of guanine residues. Crucially, longer guanine runs are thought to enhance replication primer generation (Tan et al., 2016). While it was initially suggested that the risk of reversion might be reduced by matching donor/recipient pairs according to their CSBII haplotypes (Ma et al., 2017b), a subsequent analysis indicated the incidence of reversion was not reduced in ES cell lines derived from donor/recipient pairs with the same CSBII haplotype (Hudson et al., 2019). It was therefore proposed that a strategy involving exploitation of CSBII sequence divergence might be more appropriate (Hudson et al., 2019; Kang et al., 2019). According to this, choosing a mitochondrial donor whose CSBII sequence has longer G runs than the nuclear donor (patient) may help to reduce the risk of reversion.

MtDNA reversion may also be facilitated by a founder cell effect arising from uneven distribution of maternal mtDNA in the zygote upon karyoplast fusion. Should this persist until the embryo divides to the 2-cell stage, one of the daughter cells may contain higher levels of heteroplasmy, which may be further enriched in a subset of cells during subsequent divisions due to unequal partitioning or random genetic drift. It is conceivable that the combined effect of a favorable CSBII haplotype and a founder cell effect may tilt the balance in favor of reversion. A further consideration is the potential for positive selection of some pathogenic variants (Otten et al., 2018), which would obviously have potential implications for the risk of disease transmission. So far, mtDNA reversion has been observed only in ES cell lines and it is not clear whether it could also occur during development in vivo. However, it should be noted that studies in mouse (Meirelles and Smith, 1998) and monkey (Ma et al., 2021) have reported elevated levels of heteroplasmy in some tissues. This could be explained by a founder cell effect arising from unequal distribution in the zygote cytoplasm resulting in subsequent unequal segregation of maternal mtDNA between daughter cells. Taken together, reversion in ES cells and elevated levels of heteroplasmy for maternal mtDNA in some organ systems, underscore the importance of the development of techniques to minimize the contribution of maternal mtDNA to the mitochondrial genome of the embryos (Greenfield et al., 2017).

While ongoing research is focused on improving the efficacy of MRT procedures, MST has been used in a clinical treatment in the US/Mexico to prevent transmission of the m.8993T>G mutation. The treatment resulted in a live birth with 2.36%–9.23% maternal mtDNA in tested tissues (Zhang et al., 2017). However, the use of MRT was subsequently prohibited in the US (see below). PNT has been translated to clinical treatment in the UK and is being conducted as part of a clinical study under license from the UK regulatory body, the Human Fertilization and Embryology Authority (HFEA) (Greenfield et al., 2017). Given that MRT is still at a relatively early stage of development, follow-up of any children born is of paramount importance. Moreover, until we have a more robust evidence base for MRT, it would be appropriate to offer it only to those who are unlikely to benefit

from PGT. In this regard, the mother of the baby born following MST in Mexico/US had a relatively low level (24.5%) of heteroplasmy for the m.8993T>G mutation (Zhang et al., 2017) and would therefore likely have benefited from the more established technique of PGT.

## Preimplantation genetic testing versus mitochondrial replacement therapy

While MRT is the only option for a biologically-own child in the cases where pathogenic variants are present at homoplasmic levels, the choice between PGT and MRT is less clear in cases of medium to high heteroplasmy. As mentioned above a model (Pickett et al., 2019) and an algorithm (Lindsey et al., unpublished data) should facilitate making this choice. At the moment, in the UK, MRT is offered only to women who would be unlikely to benefit from PGT. Such cases include those who carry a pathogenic variant at homoplasmic levels or at very high levels of heteroplasmy. In future, if more data on the success rate and safety of MRT become available, the situation might change and reproductive counseling might provide a choice between PGT and MRT for every heteroplasmic mtDNA mutation.

An additional advantage of PGT over MRT is that is does not require the use of donor eggs. In many countries, especially those in northern Europe, the availability of egg donors for ART is a major limitation (Bracewell-Milnes et al., 2021). The situation is less critical in some other countries, and it might in theory be possible to import vitrified oocytes for MRT treatment. However, evidence from preclinical studies indicates that the outcome of MRT is likely to be compromised by using cryopreserved (vitrified) donor eggs (Hyslop et al., 2016). By definition MRT involves creating new mixes of nuclear and mitochondrial genomes and matching donor/recipient pairs according to mtDNA haplogroups. Based on findings in laboratory mice and flies it has been suggested as a precautionary measure (Morrow et al., 2015; Røyrvik et al., 2016; Latorre-Pellicer et al., 2016). However, the extent to which matching according to haplogroup-defining variants would reduce overall mtDNA sequence divergence between donor/recipient pairs is unclear. Moreover, taking account of the difficulty obtaining donated eggs, a requirement for mtDNA haplogroup matching would severely constrain the delivery of a clinical program of MRT.

## Legal aspect of mitochondrial replacement therapy

Clinical translation of MRT also requires changes to legalization governing ART and embryo research in many countries. In 2015, the UK Parliament approved legislation to allow MRT to be used in the clinic under tight regulations. The Parliamentary vote was preceded by extensive public consultations and ethical debates. In addition, an expert panel was convened by the HFEA to consider the safety and efficacy of MST and PNT. The findings, published in three reports (http://www.hfea.gov.uk/8807.html), recommended a number of tests which it believed should be completed before MRT could be offered in clinical treatment (Greenfield et al., 2017). Once enacted, the change to UK legislation made it possible for the HFEA to consider license applications from IVF clinics to offer MRT in clinical treatment. The first such license was granted in 2018 enabling a clinical study to investigate the efficacy of PNT in clinical treatment. Under UK law, MRT can only be offered in cases where there is a risk of transmitting serious mtDNA disease. Moreover, the regulations require submission for HFEA approval for each individual case. Approval is granted only in cases where PGT is unlikely to be successful.

In the United States, the FDA started a process in 2014 to evaluate whether to approve preclinical or clinical trials using MRT (http://www.fda.gov/advisorycommittees/committeesmeetingmaterials/bloodvaccinesandotherbiologics/cellulartissueandgenetherapiesadvisorycommittee/ucm406412.htm). As part of this, the Institute of Medicine (IOM) of the National Academies of Sciences, Engineering, and Medicine was asked to assemble an expert committee to consider the ethical, social and policy issues raised by the techniques. This committee concluded that it is ethically permissible to conduct clinical investigations of MRT(s), subject to certain conditions and principles (http://www.nationalacademies.org/hmd/Reports/2016/Mitochondrial-Replacement-Techniques.aspx). However, due to federal legislation passed in December 2015, the FDA is currently unable to grant permission for any MRT efforts (http://www.sciencemag.org/news/2016/02/boys-only-panel-endorses-mitochondrial-therapy-says-start-male-embryos). Thus, in the US, restrictions on funding rather than federal laws on human embryo research hamper further developments.

Following public consultation, the Australian government has in April 2022 passed legislation (Maeve's law) to permit the use of MRT to prevent transmission of serious mtDNA disease. This required amendments to existing legislation, including lifting the prohibition on the creation of embryos for research. The findings of a public consultation on the proposed amendments were published in August 2021. In many countries, the requirement to create human embryos for research in MRT has been a major barrier to its introduction. For example, in the Netherlands, already in 2001, mitochondrial replacement therapy was considered ethically

acceptable by the Ministry of Health, if safe and in line with embryo legislation, but as in many European countries, legislation forbids the creation of embryos for research (http://wetten.overheid.nl/BWBR0013797/2013-09-27). However, the Dutch government announced the adjustment of the Embryo Act in 2016, permitting the creation of embryos for research in some specific situations, including MRT research. This means preclinical safety studies will since then be legally allowed. However, given the initially limited patient group opting for MRT and the potential risks it seems most logical to concentrate the expertise in the UK. Of note, should MRT prove to be effective and safe, clinical application of MRT would in fact be allowed following Dutch legislation, even though the 1997 Oviedo Convention (Declaration on Human Rights and Biomedicine) of the Council of Europe was signed, which includes a general prohibition of germline gene modification, the Netherlands has made a provision allowing her upon ratification to exclude the mitochondrial DNA from this ban (Hens et al., 2015). However, in other countries, like the Ukraine and Greece, more liberal legislation exists on MRT and in some countries no legislation exists at all.

## CONCLUSION

Recurrence risks and preferred reproductive option(s) for mitochondrial diseases primarily depend on the genetic cause, being either in the nuclear genome or in the mtDNA. The majority of mitochondrial diseases is caused by mutations in nuclear genes and segregate in a Mendelian way. Prenatal diagnosis (PND) or preimplantation genetic testing (PGT) are available to prevent the birth of another severely affected child. In 15-25% of cases, mitochondrial diseases are caused by mitochondrial DNA (mtDNA) mutations. For mtDNA diseases, recurrence risks and preferred reproductive option(s) additionally depend on de novo or familial occurrence, on the nature of the specific mutation, on the maternal mutation load and on national legislation. Obviously, also patient preferences are part of the decision-making, a well-informed choice being important. The severity of the disease associated with the mtDNA mutation and particularly the couple's personal experience with it in the family are likely to influence the risks a couple is willing to take versus the treatment burden they deem acceptable. At present reproductive options to pursue likely unaffected, biologically-own offspring exist for the majority of mtDNA mutation cases in the form of PND or PGT. In those cases where PND and PGT are unsuitable, for example when a pathogenic mtDNA variant is present at homoplasmic levels, MRT is the only available option for couples wishing to have a biologically-own child.

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